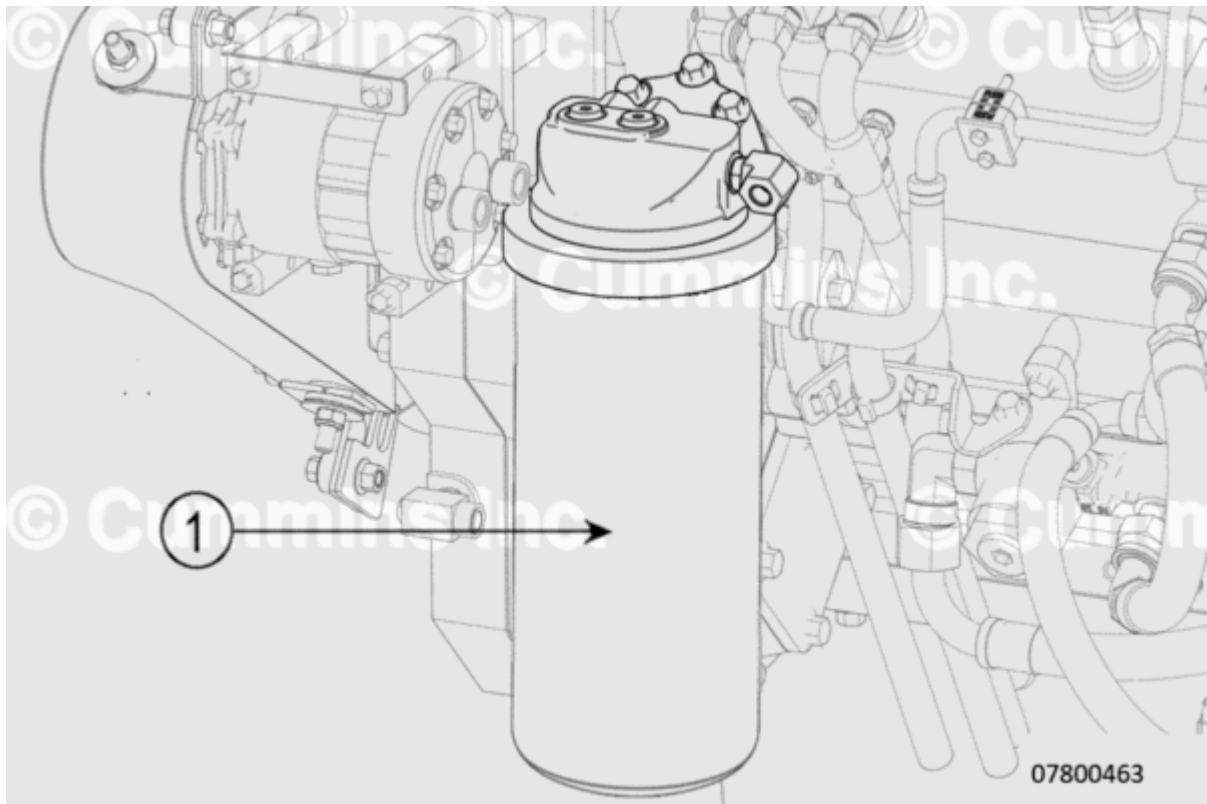


Trainings ▾

Exploded View



Fuel Pump Drive Lubricating Oil Filter Exploded View

1. Lubricating oil filter

Select Service Tools

Recommended Cummins® Service Tools

- Oil filter wrench, Part Number 3400157, or equivalent

Additional Service Items

- No additional service items required

General Information

A dedicated full flow lubricating oil filter is used to provide added protection to critical components in the high-pressure fuel pump.

The filter is mounted on a dedicated filter head on the top of the fuel pump adapter drive. Certain engines may have the filter head mounted remotely.

All components handled in this procedure weigh less than 23 kg [50 lb].

Initial Check

⚠ WARNING ⚠

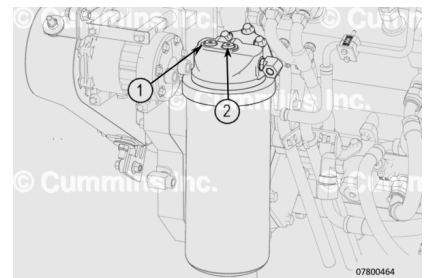
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Disconnect batteries and power supplies. See equipment manufacturer service information.

Disconnect air supply line from air starting motor, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022-tr.html>)

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Note : Some oil may drain from port in filter head when plug is removed.

Remove the Fuel Pump Oil Filter Head Outlet Pressure Port (2).

Install Compuchek™ fitting into M14 port.

Connect pressure transducer and multimeter to Compuchek™ fitting.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect batteries and power supplies. See equipment manufacturer service information.
- Connect air supply line from air starting motor, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022-tr.html>)

Connect INSITE™ electronic service tool to engine data link.

Start engine and run at idle.

Idle engine for at least 5 minutes to allow oil pressure to stabilize.

The minimum threshold for fuel pump oil pressure sensor at engine speed above 1400 rpm is 250 kPa [37 psi].

The minimum threshold for fuel pump oil pressure sensor at engine speed below 1000 rpm is 150 kPa [22 psi].

If the fuel pump oil pressure reading is below the minimum threshold at rated rpm, replace the fuel pump drive lubricating oil filter.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Shut engine OFF.

Disconnect batteries and power supplies. See equipment manufacturer service information

Disconnect air supply line from air starting motor, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022-tr.html>)

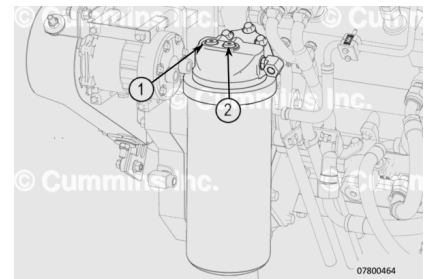
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contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



Remove Compuchek™ fitting.

Install M14 straight thread o-ring plug. Use new o-ring seal.

Torque Value: 11 n•m [96 in-lb]

Preparatory Steps

⚠ WARNING ⚠

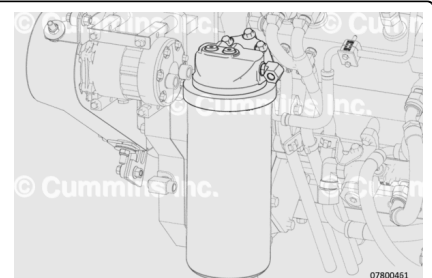
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect batteries and power supplies. See equipment manufacturer service information.
- Disconnect air supply line from air starting motor, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022-tr.html>)

Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

Remove oil filter. Use oil filter wrench, Part Number 3400157, or equivalent.

Discard filter if **not** required for analysis.

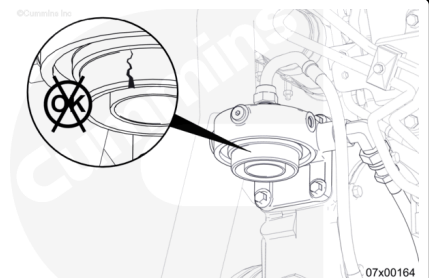
Clean and Inspect for Reuse

Clean oil filter head sealing surface. Use lint-free cloth.

Inspect oil filter head.

Replace filter head if:

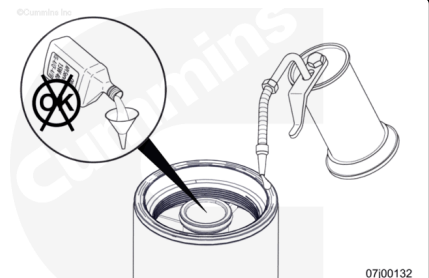
- Cracked
- Threads damaged



Install

Lubricate oil filter rubber seals. Use clean engine oil.

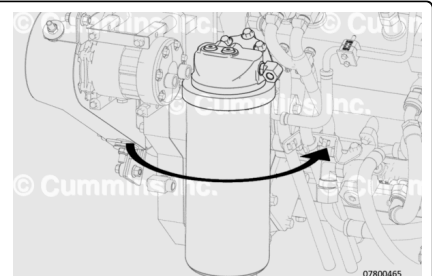
Do **not** prefill oil filter.



⚠ CAUTION ⚠

Mechanical overtightening can distort threads or damage filter element seal.

⚠ CAUTION ⚠



Locate oil filter wrench near top of oil filter canister. This will reduce possibility of damaging oil filter.

Install the oil filter. Turn until seal contacts filter head sealing surface.

Turn oil filter additional $\frac{3}{4}$ turn. Use oil filter wrench, Part Number 3400157, or equivalent.

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect air supply line to air starting motor, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022-tr.html>)
- Connect batteries and power supplies. See equipment manufacturer service information.
- Fill lubricating oil pan, if necessary. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>) in Section 7.
- Operate engine. Check for leaks.